



# The ProbLemma's Channel Season 3 Guide

## Season 3 Episode 1: The Index Of Tools

- All the problem-solving approaches studied so far, across seasons 1 and 2, are indexed by the episodes in which they were defined and practiced

## Season 3 Episode 2: An Integral Transformation

- Problem **S3M1** solved, show that for all  $\theta$ ,  $x$  real, with  $x > 1$ , and  $n = 1, 2, 3, 4, 5 \dots$  it is the case that:

$$\int_0^\pi \left( x + \sqrt{x^2 - 1} \cos(\theta) \right)^n d\theta = \int_0^\pi \frac{d\theta}{\left( x - \sqrt{x^2 - 1} \cos(\theta) \right)^{n+1}}$$

- Problems **S3M2/3/4** formulated:
  - Problems **S3M2/3/4**: deduce in at least 3 distinct ways an alternative expression (a product) for the following finite sum:

$$\sum_{k=0}^n \cos(a + k\theta)$$

## Season 3 Episode 3: One Telescope, Two Telescopes

- Problem **S3M2** solved

## Season 3 Episode 4: A Complex Approach

- Problem **S3M3** solved

## Season 3 Episode 5: Euclid Says

- Problems **S3M4** solved
- Problem **S3P1** formulated:
  - **S3P1**: find the period of oscillation of a triangular spring oscillator

## Season 3 Episode 6: Triangular Spring Oscillator

- Problem **S3P1** solved
- Problem **S3CS1** formulated:
  - **S3CS1**: sort 3 integers creatively, without the popular containers and/or industrial strength algorithms

## Season 3 Episode 7: 3-Integer Sort, A Discovery

- Problem **S3CS1** solved

- Problem S3M5 formulated:
  - S3M5: construct a point, N, inside of a right triangle, ABC, such that the magnitudes of all the angles NAB, NBC and NCA are all equal one another

#### Season 3 Episode 8: Brocard Points Introduction

- Problem S3M5 solved
- Problem S3M6 formulated:
  - S3M6: invent a way to evaluate finite sums based on a popular approach that is used in combinatorial analysis to prove binomial identities

#### Season 3 Episode 9: Summation By Double-Counting

- Problem S3M6 solved
- Problem S3P2 formulated:
  - S3P2: find the distance covered by a bouncing material point

#### Season 3 Episode 10: Infinity In Physics

- Problem S3P2 solved
- Problem S3CS2 formulated:
  - S3CS2: suggest a Turing Machine-friendly algorithm for testing if a given point is located outside or inside of a given triangle

#### Season 3 Episode 11: A Point-in-triangle Detection Algorithm

- Problem S3CS2 solved
- Problem S3M7 formulated:
  - S3M7: find a way to integrate a binomial differential  $x^m(a + bx^n)^p dx$

#### Season 3 Episode 12: Integration Of Binomial Differentials

- Problem S3M7 solved
- Problem S3M8 formulated:
  - S3M8: use the integration of binomial differentials machinery developed in the previous problem in order to evaluate the following infinite sum:

$$\sum_{n=1}^{+\infty} \frac{1}{n(n+1)(n+2)\dots(n+p)}, \quad p \geq 2$$

#### Season 3 Episode 13: Binomial Differentials Slay Infinite Sums Before Breakfast

- Problem S3M8 solved
- Problem S3P3 formulated:
  - S3P3: distinguish a sphere from an identical ball of same radius and mass

Season 3 Episode 14: Sisyphus Versus Spherical Chickens Of Uniform Density

- Problem S3P3 solved
- Problem S3CS3 formulated:
  - S3CS3: construct a single-pass algorithm for sorting 0s and 1s

Season 3 Episode 15: Single-pass Janus Sort

- Problem S3CS3 solved
- Problem S3M9 formulated:
  - S3M9: construct an equilateral triangle on 3 parallel straight lines

Season 3 Episode 16: An Equilateral Triangle On 3 Parallel Straight Lines Construction

- Problem S3M9 solved
- Problem S3M10 formulated:
  - S3M10: reconstruct a square given 4 points that belong to one side of that square each

Season 3 Episode 17: 4-Point Square Reconstruction

- Problem S3M10 solved
- Problem S3P4 formulated:
  - S3P4: find the normal and the tangential accelerations as functions of time of a point that moves along a parabola with its acceleration vector staying parallel to the  $y$ -axis at all times

Season 3 Episode 18: A Parabolic Motion With Acceleration Staying Parallel To The  $y$ -axis

- Problem S3P4 solved
- Problem S3CS4 formulated:
  - S3CS4: creatively swap 2 sub-arrays

Season 3 Episode 19: A Creative Swap Of Two Sub-Arrays

- Problem S3CS4 solved
- Problem S3M11 formulated:
  - S3M11: evaluate a finite trigonometric sum

$$\sum_{k=1}^n \frac{1}{\sin(2^k x)}$$

Season 3 Episode 20: A Finite Trigonometric Sum Evaluation

- Problem S3M11 solved
- Problem S3M12 formulated:
  - S3M12: evaluate a finite binomial sum using Physics:

$$\sum_{k=0}^n k \cdot \binom{n}{k}$$

### Season 3 Episode 21: Physics Evaluates A Finite Binomial Sum

- Problem S3M12 solved
- Problem S3P5 formulated:
  - S3P5: recover the shape of the trajectory of a point that moves in a plane in such a way that its both tangential and normal accelerations remain constant at all times

### Season 3 Episode 22: Planar Motion With Constant Tangential And Normal Accelerations

- Problem S3P5 solved
- Problem S3CS5 formulated:
  - S3CS5: explain how the Shunting Yard Algorithm works and create a Slow-Motion Player that lets a user experiment with and witness the mechanics of that algorithm

### Season 3 Episode 23: The Mechanics Of The Shunting Yard Algorithm

- Problem S3CS5 solved
- Problem S3M13 formulated:
  - S3M13: evaluate the following integrals in at least 3 different creative ways:

$$C_{a,b}(x) = \int e^{ax} \cos(bx) dx; \quad S_{a,b}(x) = \int e^{ax} \sin(bx) dx$$

### Season 3 Episode 24: 3 Creative Evaluations Of Ubiquitous Integrals

- Problem S3M13 solved
- Problem S3M14 formulated: a divisibility by 7 test
  - S3M14: design a divisibility-by-7 algorithm using the Reduce-And-Conquer approach

### Season 3 Episode 25: A Divisibility-By-7 Test Design Via Reduce-And-Conquer

- Problem S3M14 solved
- Problem S3P6 formulated: a kinematic descend of a rotating line segment
  - S3M14: recover the law of motion, the trajectory, the velocity, the acceleration and the curvature of the trajectory traced by one edge of a uniformly descending line segment that also rotates in the horizontal plane with a constant angular velocity

### Season 3 Episode 26: A Kinematic Descend Of A Rotating Line Segment

- Problem S3P6 solved
- Problem S3CS6 formulated: describe the mechanics and then design and implement the Narayana's Next-Linear-Permutation algorithm

### Season 3 Episode 27: Narayana's Next Linear Permutation Algorithm

- Problem S3CS6 solved
- Problem S3M15 formulated. Evaluate the following finite trigonometric sum:

$$\sum_{m=1}^n \arctan \left( \frac{2m}{m^4 + m^2 + 2} \right)$$

### Season 3 Episode 28: In Like A Lion Out Like A Lamb

- Problem **S3M15** solved
- Problem **S3M16** formulated. Evaluate the following limit:

$$\lim_{n \rightarrow +\infty} n^{\frac{1}{n}}$$

### Season 3 Episode 29: The Art Of Manipulation Of Inequalities

- Problem **S3M16** solved
- Problem **S3P7** formulated: deduce an equation that describes a spherical motion of a point under a constant angle with the meridians

### Season 3 Episode 30: A Deduction Of An Equation Of A Loxodrome

- Problem **S3P7** solved
- Problem **S3CS7** formulated: construct an  $O(n)$  time and  $O(1)$  space complexity algorithm for detecting if a given integer can be represented as a sum of two integers from a given array sorted in some order, ascending or descending

### Season 3 Episode 31: An Integer As A Sum Of Two Elements Of A Sorted Array

- Problem **S3CS7** solved
- Problem **S3M17** formulated: evaluate the following indefinite integral

$$\int \sqrt{x^2 - a^2} dx$$

in at least two creative ways.

### Season 3 Episode 32: Hyperbolically Speaking

- Problem **S3M17** solved
- Problem **S3M18** formulated: evaluate the following definite integrals

$$\int_0^{\frac{\pi}{2}} \sin^m(x) dx \quad \text{and} \quad \int_0^{\frac{\pi}{2}} \cos^m(x) dx$$

for  $m = 1, 2, 3, 4, 5$  and so on.

### Season 3 Episode 33: Recurrence Relations Evaluate Integrals

- Problem **S3M18** solved
- Problem **S3M19** formulated: show that hyperbolic functions can be defined by a direct analogy with circular functions defined via origin-centered unit circle

Season 3 Episode 34: Hyperbolic Functions Defined By A Direct Analogy With Circular Functions

- Problem **S3M19** solved
- Problem **S3P8** formulated: recover the law of motion of a point that is moving along a straight line that sweeps the surface of a right circular cone

Season 3 Episode 35: Twice Telescopic Rotating Antenna, The Kinematics Of

- Problem **S3P8** solved
- Problem **S3CS8** formulated: construct an algorithm for detecting a cycle in a linked list

Season 3 Episode 36: A Vicious Circle Or Lords Of The Ring

- Problem **S3CS8** solved
- Problem **S3M20** formulated: show that for all real  $x \neq \pm k\pi$ , where  $k = 0, 1, 2, 3$  and so on, it is the case that:

$$\sin(x) = x \cdot \prod_{n=1}^{+\infty} \left(1 - \frac{x^2}{n^2\pi^2}\right)$$

Season 3 Episode 37: Sine As A Product, With Applications

- Problem **S3M20** solved
- Problem **S3M21** formulated: evaluate the Euler-Poisson integral

$$\int_0^{+\infty} e^{-x^2} dx$$

using the Squeeze Theorem.

Season 3 Episode 38: What Mathematicians Do All Day Or Euler-Poisson Integral Evaluation via the Squeeze Theorem

- Problem **S3M21** solved
- Problem **S3P9** formulated: solve the **S2E54/S2P2** problem under the assumption that there is no external input into the rotating rod-bead system, which means that the system's angular velocity is no longer constant.

Season 3 Episode 39: A Bead Sliding Off Of A Massless Rod That Rotates With A Decreasing Angular Velocity

- Problem **S3P9** solved
- Problem **S3CS9** formulated: find the diamond across 8 boxes with 2 cooperating people if boxes have a random orientation, the two people must agree on the algorithm ahead of time, one person is lead into the room with the boxes to do something and then escorted out of the room, while the second person, upon entering that room, has to recover the box with the diamond

### Season 3 Episode 40: How Computer Science And Group Theory Found A Diamond

- Problem **S3CS9** solved
- Problem **S3M22** formulated: solve the following linear non-homogeneous recurrence relation:

$$t_n = 6t_{n-2} - t_{n-1} - 4, \quad t_0 = 2, t_1 = 4$$

### Season 3 Episode 41: How Generating Functions Can Solve Linear Non-Homogeneous Recurrence Relations

- Problem **S3M22** solved
- Problem **S3M23** formulated: how many chords are formed by  $n$  distinct points located on the circumference of a circle and why?

### Season 3 Episode 42: Chords Of The Ring

- Problem **S3M23** solved
- Problem **S3P10** formulated: two points execute a coplanar motion as follows. A point  $R$  moves along a straight line with constant velocity  $\vec{v}_r$ , while a point  $C$  moves (in the same plane) with a velocity,  $\vec{v}_c$ , that is constant by magnitude but always points at the point  $R$ . Find the trajectory of the point  $C$ , the condition and the time of the intercept of  $R$  by  $C$ , the intercept time from the moment when the vectors  $\vec{v}_r$  and  $\vec{v}_c$  were perpendicular and the distance  $|CR|$  was  $d_0$ .

### Season 3 Episode 43: The Kinematics Of Wile E. Coyote and Road Runner

- Problem **S3P10** solved
- Problem **S3CS10** formulated: invent/discover/construct/synthesize/put together/come up with a practically useful  $O(1)$  time complexity algorithm.

### Season 3 Episode 44: A Story Of An $O(1)$ Bucket List

- Problem **S3CS10** solved
- Problem **S3M24** formulated: solve the equation  $z \cdot z = \bar{z}$  over  $\mathbb{C}$

### Season 3 Episode 45: There Is More To Mathematics Than Meets They Eye

- Problem **S3M24** solved
- Problem **S3M25** formulated: find the shape of an orbit from each point of which both the Earth and the Moon are seen under one and the same angle

### Season 3 Episode 46: Where Problems Live Or Earth-Moon Equiangular Orbit

- Problem **S3M25** solved
- Problem **S3P11** formulated: find the velocity of a point that orbits a circle in such a way that its areal velocity with respect to an off-center point located strictly inside of that circle is constant as a function of the distance from that off-center point to the moving point itself

### Season 3 Episode 47: A Hidden Kepler Or Isomorphism In Celestial Mechanics

- Problem **S3P11** solved

- Problem **S3CS11** formulated: implement a thread-safe blocking  $O(1)$  queue following the spirit of the solution of the problem **S3CS10**, ensuring that this queue exhibits the following behavior:
  - it blocks if it is full on a call to `qAdd()` and
  - it blocks if it is empty on a call to `qRm()`

#### Season 3 Episode 48: A Thread-Safe Blocking $O(1)$ Add/Remove Queue

- Problem **S3CS11** solved
- Problem **S3M26** formulated: show that no non-trivial harmonic sum  $H_{n>1}$ :

$$H_n = \sum_{k=1}^n \frac{1}{k}$$

is ever an integer.

#### Season 3 Episode 49: A Harmonic Argument

- Problem **S3M26** solved
- Problem **S3M27** formulated: explain why  $\zeta(1)$  diverges but  $\zeta(2)$  converges.

#### Season 3 Episode 50: Why $\zeta(1)$ Diverges But $\zeta(2)$ Converges

- Problem **S3M27** solved
- Problem **S3P12** formulated: determine the velocity and the acceleration of a point moving over an origin-centered ellipse, with semi-axes  $a$  and  $b$ , with a constant velocity,  $v_0$ , as functions of the point's coordinates,  $x$  and  $y$ .

#### Season 3 Episode 51: A Uniform Motion Over An Ellipse, A Verbal Exam Scenario

- Problem **S3P12** solved
- Problem **S3CS12** formulated: construct an  $O(n)$  algorithm for reversing the direction of a singly-linked list to its opposite in a single pass.

#### Season 3 Episode 52: Hansel And Gretel Do Comp Sci

- Problem **S3CS12** solved
- Problem **S3M28** formulated: decompose the following real-to-real single-variable functions:

$$\cot(x), \quad \tan(x), \quad \frac{1}{\sin(x)}, \quad \frac{1}{\sin^2(x)}$$

into the corresponding partial fraction series using the machinery of Real Analysis only.

#### Season 3 Episode 53: A Real Cornucopia Of Partial Fraction Decompositions

- Problem **S3M28** solved
- Problem **S3M29** formulated: evaluate the following improper integral:

$$\int_0^{+\infty} \frac{\sin(x)}{x} dx$$

using the decompositions obtained in [S3M28/S3E53](#).

Season 3 Episode 54: Partial Fraction Decompositions Evaluate Integrals

- Problem [S3M29](#) solved
- Problem [S3P13](#) formulated: determine the trajectory of a ladybug sitting on a rake that slides down a barn floor in such a way that the top of the rake's handle remains in contact with the barn's wall at all times.

Season 3 Episode 55: Kinematics In A Barn Or A Ladybug On A Rake

- Problem [S3P13](#) solved
- Problem [S3CS13](#) formulated: show

Season 3 Episode 56: Strip Teeze

- Problem [S3CS13](#) solved
- Problem [S3M30](#) formulated: prove the relevant theorems for the Frullani Integrals

$$\int_0^{+\infty} \frac{f(ax) - f(bx)}{x} dx$$

Season 3 Episode 57: Frullani Integrals Forever

- Problem [S3M30](#) solved
- Problem [S3M31](#) formulated: show how improper integrals of rational functions can be evaluated

$$\int_{-\infty}^{+\infty} \frac{P_m(x)}{Q_n(x)} dx, \quad m + 2 \leq n$$

Season 3 Episode 58: The Real Granddaddy of the Cauchy's Residue Theorem

- Problem [S3M31](#) solved
- Problem [S3PS14](#) formulated: find the trajectory of a point on the circumference of a circle of radius  $r$  rolling without sliding and spinning over 1) the exterior and 2) the interior of a fixed circle of radius  $R$

Season 3 Episode 59: Epi/Hypo/Cycloids, Copernicus, Ladybug, Chicken & A Vacuum Cleaner

- Problem [S3P14](#) solved
- Problem [S3CS14](#) formulated: describe the structure of the fixed points from the Strip Teeze computer science problem [S3CS13](#)

Season 3 Episode 60: The Structure Of Fixed Points Of Strip Teeze

- Problem **S3CS14** solved
- Problem **S3M32** formulated: establish the less-than/equal-to/greater-than relationship between the following two single-variable real-to-real functions,  $\cos(\sin(x))$  and  $\sin(\cos(x))$

Season 3 Episode 61: Lorem Ipsum

- Problem **S3M32** solved
- Problem **S3M33** formulated: prove without using integration:

$$\pi = 2 \cdot \int_{-1}^1 \sqrt{1-x^2} dx$$

Season 3 Episode 62: Lorem Ipsum

- Problem **S3M33** solved
- Problem **S3P15** formulated: show

Season 3 Episode 63: Lorem Ipsum

- Problem **S3P15** solved
- Problem **S3CS15** formulated: show

Season 3 Episode 64: Lorem Ipsum

- Problem **S3P15** solved
- Problem **S3M34** formulated: show

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